

FILE UNDER MYSTERY

Complete Technical Architecture & 4-Hour Event Master Plan

1. Event Structure & 4-Hour Time Distribution

To provide a rigorous, deeply engaging 3.5 to 4-hour competition (e.g., 11:30 AM to 3:45 PM), File Under Mystery is organized into **Three Investigative Phases (4 Levels each = 12 total)** culminating in a **Tri-Phased Blackbox Meta-Climax**. Teams of 3–4 students experience continuous pacing without bottlenecking.

Phase & Time Block	Levels Included	Target Duration	Forensic Domains Covered
Phase I (12:00 - 12:50)	Levels 01, 02, 03, 04	50 Minutes	Image Histogram, Audio STFT, Video Diff, LSB Stego
Phase II (12:50 - 01:50)	Levels 05, 06, 07, 08	60 Minutes	Prime Ciphers, Packet Triage, Matrix Scramble, 2D FFT
Phase III (01:50 - 02:50)	Levels 09, 10, 11, 12	60 Minutes	Elliptic Curves, Rule 30 Automata, Audio Phase, Graph Theory
Phase IV (02:50 - 03:35)	Tri-Phased Final Boss	45 Minutes	12-Token Matrix Reassembly + Live Terminal Uplink Decryption

2. Master 12-Level Forensic & Mathematical Specifications

#	Title	Technique / Math Principle	Flag	Pts
01	The Photograph	Histogram dynamic range stretch [10,40] -> [0,255]	A19X7	10
02	The Voicemail	STFT Spectrogram: 3 kHz Morse carrier wave	K4P82	12
03	The Recording	Temporal Frame Differencing $ f_t - f_{t-1} $ at Frame 142	XT4Q1	14
04	The Holiday Photo	Blue channel bitplane 0 LSB -> ZIP -> Linear Equation	M77RB	16
05	The Numbers	Prime-indexed modular shift stream: $P_n \bmod 26$	P0W3R	18
06	The Capture	HTTP log triage: Oversized Authorization Base32 token	NT2K5	15
07	The Corrupted Image	8x8 Block Permutation matrix inversion + Transpose + 90 deg	BXZ19	18
08	The Signal	2D Fast Fourier Transform radial bandpass ring filter + IFFT	FIN4L	20
09	The Trajectory	Elliptic Curve point multiplication on $y^2 = x^3 + 7 \bmod 101$	EL7P9	22
10	Lattice Growth	Rule 30 1D Cellular Automaton reverse-time seed extraction	R30S4	22
11	Dual Transmission	Binaural Phase Inversion: 180 deg channel cancellation (L - R)	PH4Z3	24
12	Campus Grid	Graph Theory: Eulerian path through prime-degree sensor nodes	GR4PH	25
BOS	The Uplink	12-Token Reverse Synthesis -> 60-char Vigenere Decryption	BEACON	40

3. Full Tech Stack & Implementation Security

- **Client-Side Stack:** React 18 + Vite + Tailwind CSS + Zustand + Lucide Icons.
- **Signal & Audio Engines:** wavesurfer.js (waveform scrubbing), fft.js (2D FFT in Web Workers), Web Audio API (Phase cancellation).
- **Mathematical Solvers:** mathjs (matrix transformations & modular arithmetic), fflate (in-memory ZIP extraction).
- **Backend & Zero-Trust Verification:** Supabase PostgreSQL + Edge Functions with SHA-256 server-side checking (no answers in client bundles).
- **Asset Generation Pipeline:** Python 3.10 (Pillow, NumPy, SciPy, Matplotlib) prebakes all 12 assets offline.